Thursday, 08 Jan 2026

Knowledge Systems Working Group

Business & Primer Working Meeting

Robert Nilsson, Bill Hubbard

INCOSE

International Workshop 2026

Knowledge Systems Working Group

Robert Nilsson, Bill Hubbard

KSWG Meeting Schedule

- ❑ ✓ Sunday, 1 Feb 2026, - Business Meeting 90 min
- ❑ Monday, 2 Feb 2026 - Primer Review Meeting 120 min
- ❑ ✓ Tuesday, 3 Feb 2026 - Business & Joint WG Meeting 120 min

KSWG Meeting Agenda

1 Feb 2026, Business Meeting, 15:30-17:00 PST

- ❑ Introductions (15 min)
- ❑ Knowledge Systems Overview, Scope, and Definitions (15 min)
- ❑ Knowledge Systems Taxonomy (15 min)
- ❑ Future State Vision / Role of AI (15 min)
- ❑ Working Group Cross-Connections (10 min)
- ❑ Primer Intro (10 min)
- ❑ Wrap up, closing comments, actions (10 min)

KSWG Meeting Agenda

1 Feb 2026, Business Meeting, 15:30-17:00 PST

Future State Vision / Role of AI

(15 min)

KSWG Meeting Agenda

1 Feb 2026, Business Meeting, 15:30-17:00 PST

Future State Vision / Role of AI (15 min)

- ❑ Future State Vision: Semantic Interoperability for INCOSE
  - ❑ Unified Knowledge Layer - harmonizing ontologies, taxonomies, and vocabularies across all INCOSE Working Groups
  - ❑ AI-Ready Knowledge Structures - knowledge organized so both humans and AI systems can reliably interpret it
  - ❑ Democratized Knowledge Access - self-service discovery while maintaining governance

KSWG Meeting Agenda

1 Feb 2026, Business Meeting, 15:30-17:00 PST

Future State Vision / Role of AI (15 min)

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Conceptual Architecture

Stakeholder Layer
Executives, Managers, Analysts, Engineers

Application Layer
Queries, Dashboards, Reports, AI Assistants

Semantic Knowledge Layer
Core Ontology, Domain Ontologies, Controlled Vocabularies

Data Source Layer
PLM, ERP, CRM, Documents, External Sources

KSWG Meeting Agenda

1 Feb 2026, Business Meeting, 15:30-17:00 PST

Future State Vision / Role of AI (15 min)

- Future State Vision: Addressing Working Group Heterogeneity
  - Common Semantic Patterns Across Disciplines - identifying shared conceptual patterns while preserving local context
  - Federated Governance Model - autonomy for each WG while contributing to shared infrastructure
  - Progressive Integration Pathways - incremental adoption through high-value use cases

KSWG Meeting Agenda

1 Feb 2026, Business Meeting, 15:30-17:00 PST

Future State Vision / Role of AI (15 min)

Role of AI: A Secondary Benefit of Good Semantic Architecture

Primary benefit: Human comprehension of human knowledge across engineering disciplines.

- Context-Aware AI Systems - well-structured knowledge provides grounding for AI queries
- Semantic Constraints for AI Outputs - ontological structure improves AI reliability and verifiability
- Human-AI Collaboration Patterns - explainable AI through traceable semantic relationships
- Combining AI with Traditional Analysis - AI augments but doesn't replace domain expertise

KSWG Meeting Agenda

1 Feb 2026, Business Meeting, 15:30-17:00 PST

Future State Vision / Role of AI (15 min)

Conceptual Flow (not a production architecture)

Input Sources	Semantic Layer	Output Capabilities
Engineer Query	Controlled Vocabulary	Verified Responses
Requirements	Domain Ontology	Explainable Paths
Architecture Models	Relationship Graph	Impact Analysis

KSWG Meeting Agenda

2 Feb 2026, Business Meeting, *(to be scheduled)*

- ❑ Introductions (10 min)
- ❑ Knowledge Systems Overview, Scope, and Definitions (10 min)
- ❑ Ontologies in practice (15 min)
- ❑ Semantic Interoperability (15 min)
  - Relationships between SysML v2, JSON-LD, RDF, LPG, and UAF
- ❑ Primer Review (60 min)
- ❑ Wrap up, closing comments, actions (10 min)

KSWG Meeting Agenda

3 Feb 2026, Business Meeting & Joint WG Meeting, *10:00-12:00 PST*

- ❑ Introductions (10 min)
- ❑ Knowledge Systems Overview, Scope, and Definitions (15 min)
- ❑ Systems Science Overview (15 min)
- ❑ Common fundamentals between Working Groups & future work (60 min)
- ❑ Revisit Knowledge Systems / System Science Taxonomy (10 min)
- ❑ Primer Edit (0 min)
- ❑ Wrap up, closing comments, actions (10 min)

KSWG Meeting Agenda

3 Feb 2026, Business Meeting & Joint WG Meeting, *10:00-12:00 PST*

Common fundamentals between

Working Groups & future work

(60 min)

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KSWG Meeting Agenda

1 Feb 2026, 15:30-17:00 PST

Foundational Interoperability (The Digital Core)

Common fundamentals between Working Groups & future work (60 min)

- The Urgency of Semantic Interoperability
  - Current Challenge: INCOSE Working Groups often operate in silos, creating "static models" that cannot be easily shared or understood across different engineering tools.
  - The Risk of Inaction: Without a shared technical "language," we face manual translation errors and a lack of verifiable outputs for complex systems.
  - Our Opportunity: Transition from static documentation to a Unified Knowledge Fabric that harmonizes ontologies and vocabularies across all of INCOSE

Bridging the Gap Between Engineering Disciplines

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KSWG Meeting Agenda

1 Feb 2026, 15:30-17:00 PST

Foundational Interoperability (The Digital Core)

Common fundamentals between Working Groups & future work (60 min)

- ❑ Enable “Translation” Across Systems Engineering Disciplines
  - ❑ MBSE & Digital Engineering: Convert your static models into dynamic, machine-understandable knowledge graphs.
  - ❑ AI Systems & Requirements: Establish the ontological "guardrails" needed to ensure AI-generated requirements are verifiable.
  - ❑ Security & Resilience: Create an organizational "Immune System" where threat detection automatically updates system parameters across the enterprise.

Why Every Working Group Needs a Knowledge Graph

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KSWG Meeting Agenda

1 Feb 2026, 15:30-17:00 PST

Foundational Interoperability (The Digital Core)

Common fundamentals between Working Groups & future work (60 min)

- Future State Vision: Democratized Knowledge Access
  - AI-Ready Infrastructure: We are structuring knowledge graphs specifically for GenAI integration through RAG (Retrieval-Augmented Generation) architectures.
  - Context-Aware Systems: Knowledge graphs provide the rich contextual grounding necessary for GenAI to assist in complex engineering tasks.
  - Federated Governance: Each Working Group maintains autonomy over its domain specialty while contributing to a shared infrastructure

The Role of AI in Scaling Engineering Expertise

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KSWG Meeting Agenda

1 Feb 2026, 15:30-17:00 PST

Foundational Interoperability (The Digital Core)

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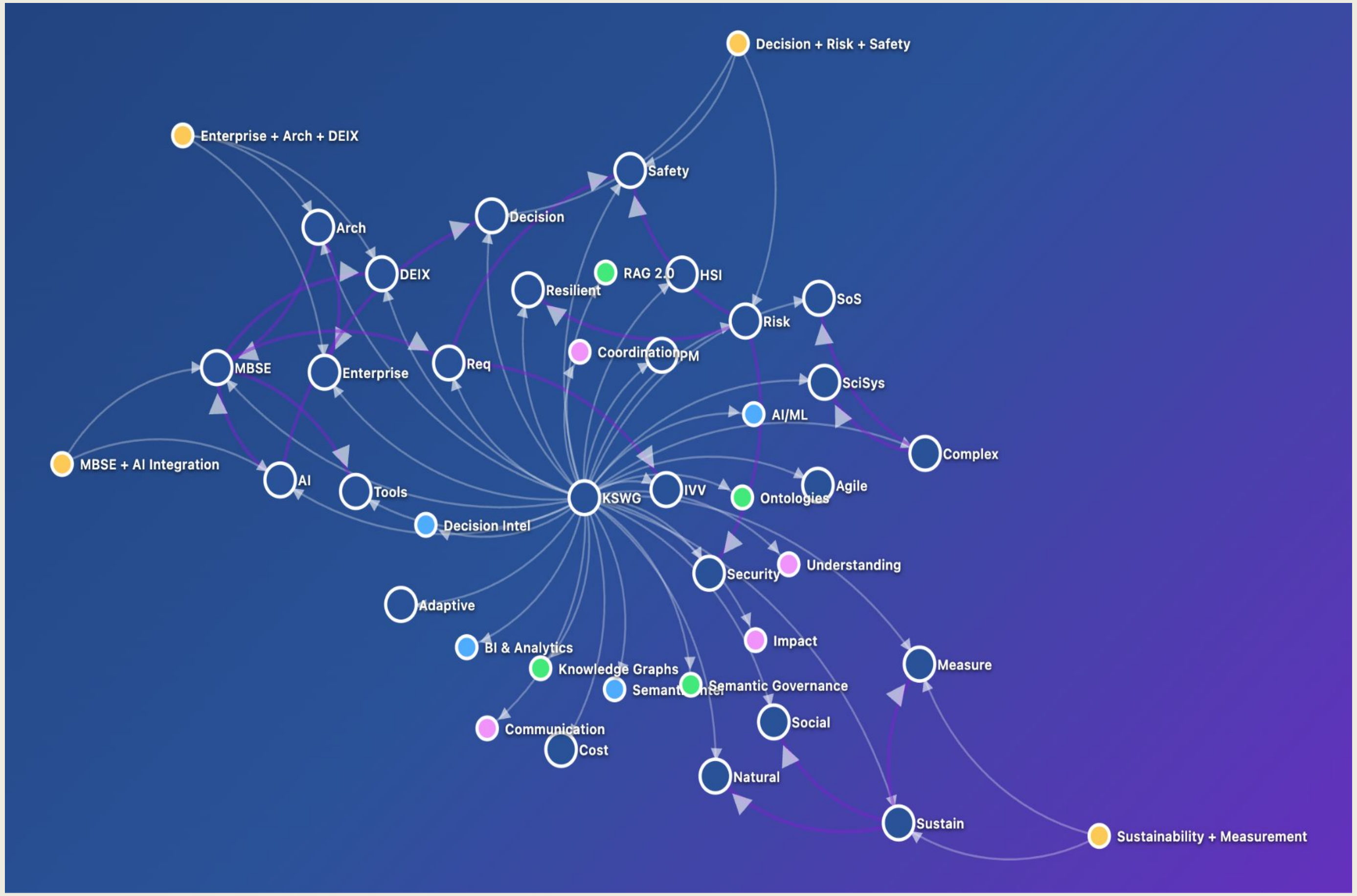
Common fundamentals between Working Groups & future work

30 Min. Exercise:
Map Your Domain to the Global Fabric



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KSWG Meeting Agenda

1 Feb 2026, 15:30-17:00 PST

Foundational Interoperability (The Digital Core)

Common fundamentals between Working Groups & future work (60 min)

- Interactive Exercise: "The Interoperability Challenge"
  - Identify Your "Core Terms" (10 min):
    - List 3 critical concepts or requirements from your Working Group (e.g., "Risk Level," "Safety Protocol," "Sustainability Constraint").
  - Mapping the Connection (10 min):
    - Partner with a member from a different Working Group.
    - How does your "Risk Level" impact their "Project Schedule" or "Human Workflow"?
  - Define the Guardrail (10 min):
    - What is one "machine-readable constraint" that must exist to ensure an AI doesn't misinterpret your data when passing it to another group?

30 Min. Exercise: Objective: Map Your Domain to the Global Fabric

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Todays Agenda

- Agenda items
  - Future meeting schedule.
  - International Workshop 2026 in Torrance California
  - **Working Group Collaborations**
  - Open Mic; Controlled Vocabulary, Semantics, Graph Analytics, Tools, AI4KS

INCOSE's International Workshop is the event of the year for systems engineers to contribute to their ind…

What Working Groups do we want to collaborate with?

Attendees spend four focused days working alongside fellow syste… a difference. Unlike INCOSE's annual **International Symposium** and othe… paper, panel, or tutorial presentations. Systems Engineers at all levels … all backgrounds are encouraged to engage in working sessions, and contribute their k…edge and experience to take the discipline forward.

Sessions are encouraged to inclusively engage multiple working groups, diverse practitioners, and multiple competency levels as they complete meeting outcomes, satisfying the session's purpose. Attendees should experience a different yet complementary experience to regular working group meetings throughout the year.

IW 2026 Attendance

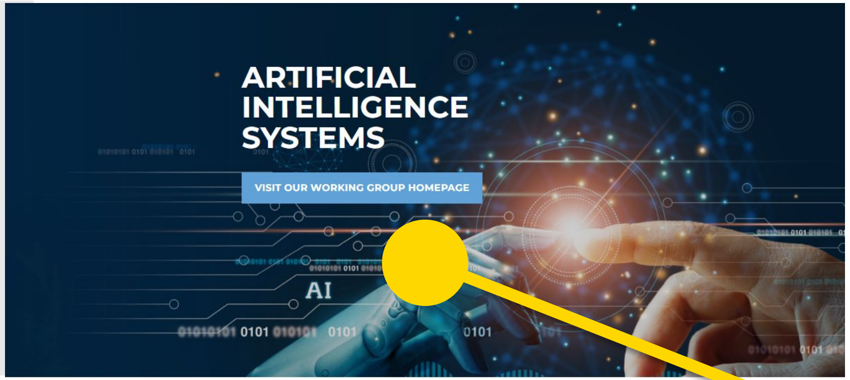
IW 2026 is open to **INCOSE** members, and non-members are encouraged to join **INCOSE** prior to the IW. For more information about membership, please visit **https://www.incose.org/incose-member-resources/join-incose** or email **info@incose.net**.

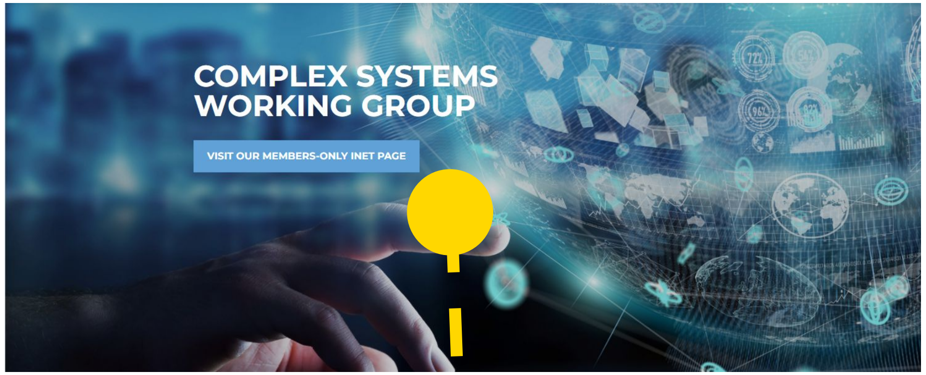
Working Group Collaborations

Artificial Intelligence Systems

Transformational

Ali K. Raz, Chair
Ramakrishnan "Ramki" Raman, Co-Chair





Complex Systems

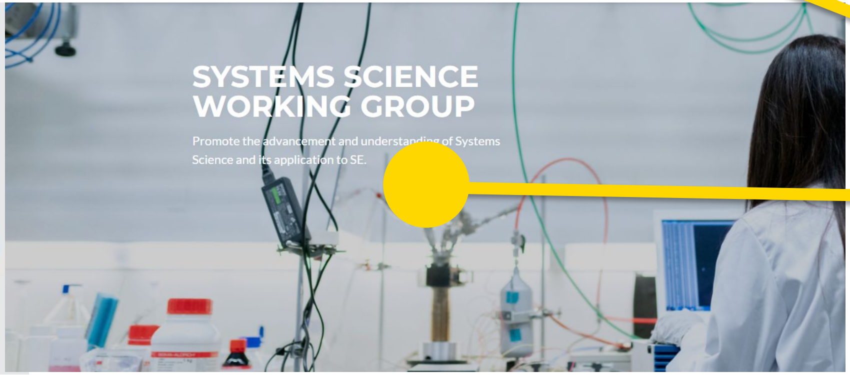
Analytic Enablers

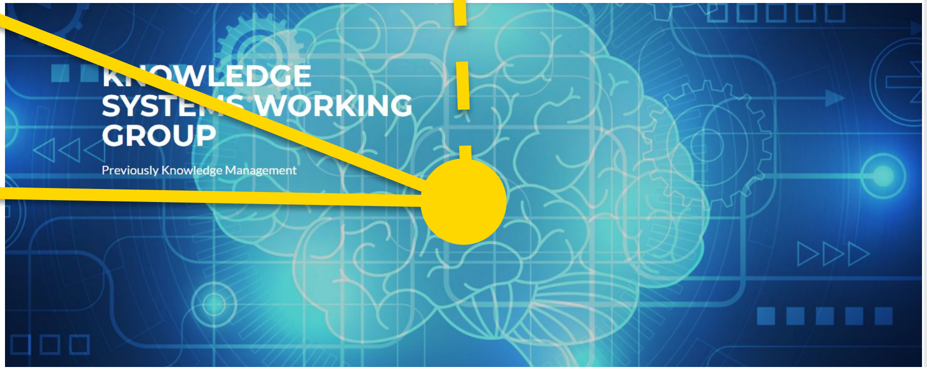
Dean Beale, Chair
Ali K. Raz, Co-Chair
Andrew C. "Andy" Pickard, Co-Chair
Michael D. Watson, Co-Chair

Systems Science

Transformational

Javier Calvo-Amodio, Chair
Harington Lee, Co-Chair
James N. Martin, Co-Chair





Knowledge Systems

Transformational

Robert Nilsson, Co-Chair
William Hubbard, Co-Chair

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  - **Open Mic; Controlled Vocabulary, Semantics, Graph Analytics, Tools, AI4KS**

INCOSE Knowledge Systems WG